MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SCHOOL OF COMPUTER ENGINEERING

V SEMESTER B.TECH. (CCE)

ICT 3128 Wireless Communication and Computing

Date: 29-08-2026 FISAC-1: Group Assignment Marks: 05

Assignment Submission Instructions

1. **Assignment Details:** Mention the following details on the assignment sheet:
   - Subject Code
   - Subject Name
   - Date
   - Section
   - Group Number
   - Registration Number of each group member
   - Names of all group members

2. **Contribution Details:** The assignment sheet must include a **table showing the contribution of each group member**.
3. **Experiment Files:** All files used for the experiment must be placed in **one ZIP folder**.
4. **Submission Format:** Upload **only one PDF file** and **one ZIP folder**. (Only Team lead has to upload)
   - PDF file name: **GroupNumber_QuestionNumber**
   - Example: **G01_Q2.pdf**
   - The ZIP folder should contain all the files used for the experiment.

5. **Last Date for Submission: 01/11/2026 (1 November 2026), on or before 5:00 PM**

Important: Please ensure that the submission follows the above format. Submissions that do not follow the specified format may not be considered.

MATLAB Software: Meet Lab instruction Mrs. Tulasi, Computing Lab -15, 1st Floor, SCE, AB-5, MIT

Common Project Requirements

- Every question has the same workload: 4 transmitter tasks, 2 channel/propagation tasks, 2 receiver tasks and 3 analysis/engineering-decision tasks.
- All projects must implement or explicitly model a complete transmitter - channel/environment - receiver chain.
- Use a reproducible payload; where registration-number-derived data is requested, document the conversion and framing method.
- Include quantitative validation using appropriate metrics such as BER, PER, EVM, SINR, outage, throughput, latency, capacity, spectral efficiency or link margin.
- Where Mobile apps/sniffers are used, analyze only permitted measurements and the group's own traffic and follow institutional/network policies.

Mobile APP		**Mobile APP**
Network signal guru		CellMapper
G-NetWiFi		NetMonitor cell Signal Logging
Network Cell Info		DW-RF Mobile Trace
Network Signal Info lite		WiFi Analyzer
NetMonster		WiFi Monitor network analyzer
LTE Discovery (5G NR)		Cell Signal Monitor
Network survey		RF Signal Tracker and Detector
4G LTE/5G Coverage monitor		Signal Detector (lefan Co. Ltd)
Cellular-Z		G-Net Tracker Lite
Analiti		Tower Collector

- Each project must end with a justified engineering decision, not only plots or screenshots.

20. Integrated Wireless Capstone: Adaptive Cellular/WLAN Transceiver with Coverage, Mobility and Capacity Validation

BTL6 - Create

Syllabus: Wireless system design; Link budget; Fading; Cellular concepts; OFDM/MIMO; WLAN; Coverage and capacity

Project scenario: Create an integrated end-to-end wireless prototype that combines a realistic transmitter, channel/environment and receiver and uses at least one field-measurement input from Mobile Apps (page-2). The project shall demonstrate how physical-layer design, coverage and user performance interact in one complete system.

A. Transmitter design and implementation

- Generate framed traffic with preamble/payload/CRC and implement an OFDM-based transmitter using adaptive QPSK/16-QAM/64-QAM or a justified equivalent.
- Include configurable transmit power, antenna gain, channel bandwidth and optional 2x2 MIMO mode.
- Use one permitted field-measurement dataset to calibrate or validate a link parameter such as path loss, RSSI trend, channel occupancy or latency.
- Plot transmitter spectrum/resource mapping and calculate nominal throughput and EIRP/link parameters.

B. Channel / propagation model

- Model path loss, shadowing, multipath fading, AWGN and one additional impairment such as interference or mobility/Doppler.
- Evaluate a scenario containing at least three receiver locations or mobility states and two traffic/load conditions.

C. Receiver design and implementation

- Implement synchronization, channel estimation/equalization, adaptive demodulation and CRC/PER measurement; include MIMO detection if selected.
- Recover the payload and log received power/SINR, BER/PER, throughput and latency/outage proxy across the scenario.

D. Analysis, validation and engineering decision

- Validate at least one simulated trend against the mobile apps (page-2) measurement and explain deviations.
- Identify the dominant limitation among coverage, interference, mobility, bandwidth and receiver performance.
- Produce a final engineering configuration and justify it against explicit reliability, throughput and coverage targets.

Required project evidence: Complete MATLAB/Simulink project, end-to-end architecture, field evidence, parameter/link-budget tables, Tx/Rx plots, BER/PER/throughput/coverage results and final engineering design justification.